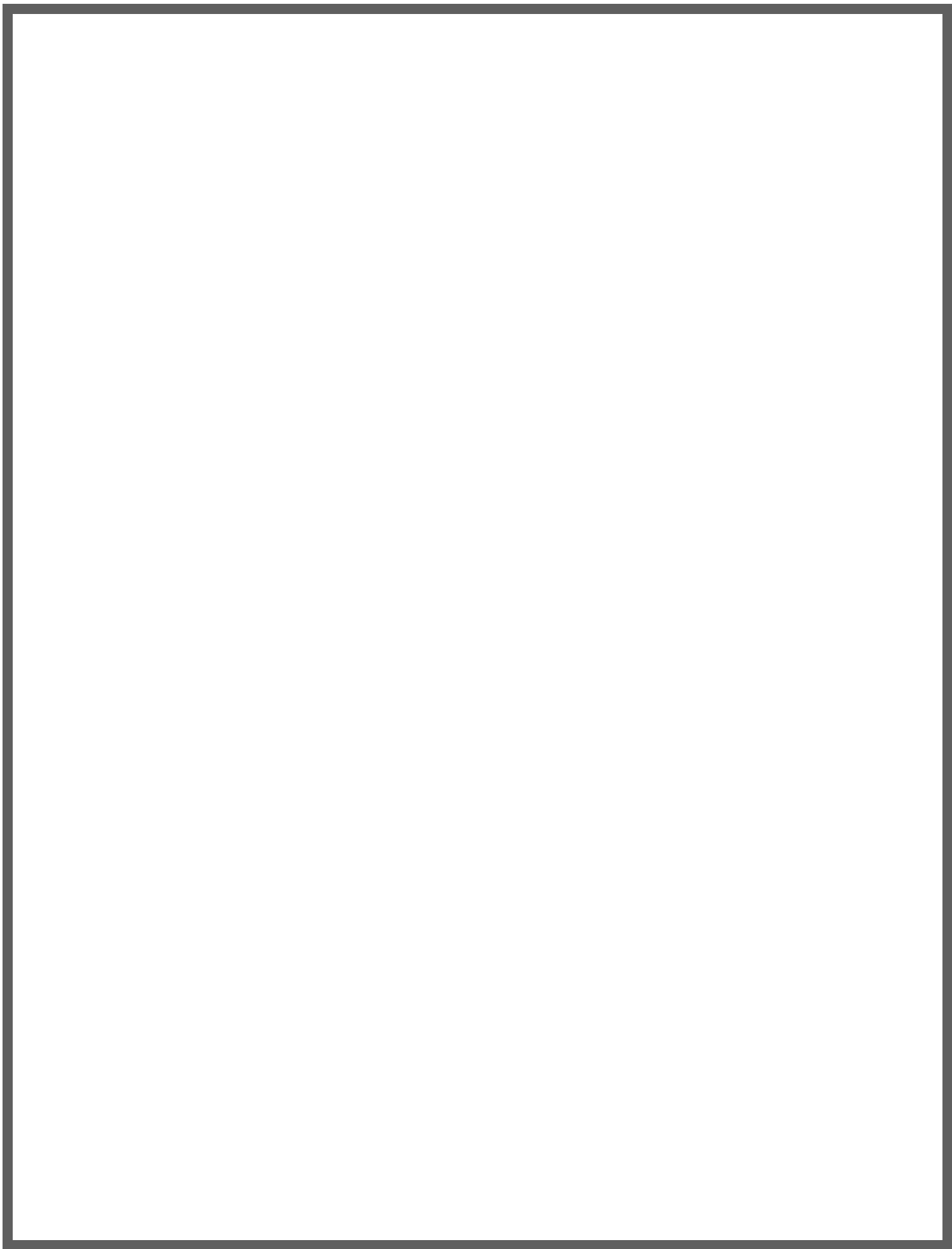




FLIGHT CENTER, LLC

**PIPER AZTEC
PA23-250
TRAINING SUPPLEMENT**



MULTI-ENGINE

AIRMEN CERTIFICATION STANDARDS

ADDITIONAL RATING TASKS

I. Preflight Preparation

- F. Performance and Limitations
- G. Operation of Systems

II. Preflight Procedures

- A. Preflight Assessment
- B. Flight Deck Management
- C. Engine Starting
- D. Taxiing
- F. Before Takeoff Check

IV. Takeoffs, Landings, and Go-Arounds

- A. Normal Takeoff and Climb
- B. Normal Approach and Landing
- E. Short Field Takeoff and Maximum Performance Climb
- F. Short Field Approach and Landing

V. Performance Maneuvers

- A. Steep Turns

VII. Slow Flight & Stalls

- A. Maneuvering During Slow Flight
- B. Power-Off Stalls
- C. Power-On Stalls
- D. Accelerated Stalls
- E. Spin Awareness

IX. Emergency Operations

- E. Engine Failure During Takeoff Before V_{MC} (*Simulated*)
- F. Engine Failure After Liftoff (*Simulated*)
- G. Approach and Landing with an Inoperative Engine (*Simulated*)

X. Multi-Engine Operations

- A. Maneuvering with One Engine Inoperative
- B. V_{MC} Demonstration
- C. One Engine Inoperative (*Simulated*) (*solely by reference to Instruments*) During Straight-and-Level Flight and Turns *
- D. Instrument Approach and Landing with an Inoperative Engine (*Simulated*) (*solely by reference to Instruments*) *

* **Commercial MEL only**

Piper Aztec Systems

We train in two models of the Piper Aztec: the “E” and the “F” models. For standardization purposes, we will use E-model procedures for all aircraft. One of our F-models has the Robertson STOL kit installed making this a very capable short field aircraft. However, we will be using the non-STOL procedures and numbers since using them will only apply an additional margin of safety to the STOL F-model. Additionally, this will enable us to standardize our procedures and speeds across our fleet of aircraft.

E-Model

- Uses MPH for all reference speeds since these aircraft incorporate an airspeed indicator in MPH.
- Uses a down-spring for the stabilator to counter act any nose-up pitching moment caused by a quick, smooth application of full power.
- Incorporates an electrical paralleling system which utilizes two 14V alternators and a single 12V battery.
- Fuel system consists of 4 integral tanks (2 inboard and 2 outboard) consisting of a capacity of 36 gallons each. This means the total capacity of the E-model is 144 gallons with 4 gallons unusable.
- Utilizes only one hydraulic pump which is installed as a left (critical) engine accessory.
- Has brakes only on the left side.

F-Model

- Uses KIAS for all reference speeds since these aircraft incorporate an airspeed indicator in KIAS.
- Incorporates a 28V system with two 28V alternators and a single 24V battery.
- Fuel system consists of 4 integral tanks (2 inboard and 2 outboard) consisting of a capacity of 36 gallons each. This means the total capacity of the E-model is 144 gallons with approximately 6.8 gallons unusable.
- The STOL F-model utilizes two hydraulic pumps; one installed on each engine. However, the non-STOL F-model Utilizes only one hydraulic pump which is installed as a left (critical) engine accessory.
- Has brakes on both sides.

Aircraft Speeds

V SPEED	E-Model MPH	F-Model KIAS	DESCRIPTION	NOTE
V _{SO}	70	55	Stall Speed (Gear down; Flaps full)	
V _S	76	61	Stall speed flaps up (Gear up; Flaps up)	
V _{MC}	80	64 (60 STOL)	Single Engine Minimum Control Speed	<i>Red Line</i>
V _R	80	70	Rotation Speed	
V _{XSE}	97	83	Best Angle Climb (Single Engine)	
V _{YSE}	102	88	Best Rate Climb (Single Engine)	<i>Blue Line</i>
V _{SSE}	N/A	80	Safe Single Engine Speed	
V _X	107	89	Best Angle Climb	<i>Short Field Takeoff Initial Climb</i>
V _Y	120	102	Best Rate Climb	
V _{A-4000 LBS}	131	115	Maneuvering Speed (4000 lbs.)	
V _{A-5200 LBS}	149	131	Maneuvering Speed (5200 lbs.)	
V _{FE-FULL}	125	108	Flaps Full	
V _{FE-1/2}	140	123	Flaps ½	
V _{FE-1/4}	160	141	Flaps ¼	
V _{LO}	150	132	Landing Gear Operation	
V _{LE}	150	132	Landing Gear Extension	
V _{NO}	198	175	Maximum Structural Cruising Speed	
V _{NE}	249	221	Never Exceed Speed	
V _{APP Flaps 1/4}	120	120	Approach Speed	<i>Flaps ¼</i>
V _{APP Flaps 1/2}	110	110	Approach Speed	<i>Flaps ½</i>
V _{APP Flaps FULL}	100	100	Approach Speed	<i>Flaps Full</i>
V _{BARRIER}	85-90	85-90 (75-80 STOL)	Barrier Speed	<i>Flaps Full</i>

One Engine Inoperative

The following list which we have dubbed ***“The Drill”*** must be committed to memory to such a degree that it is as easily recalled as *“Mary Had a Little Lamb.”* This drill should not be rushed but instead it is to be performed in a very methodical and purposeful manner so as not to skip over and/or miss any single item on the list. This procedure should be “chair flown” often to help it become part of muscle memory. Special emphasis is placed on identifying the correct prop and throttle levers before any shutdown or simulated shutdown of an engine. A three-way exchange of verification will be used.

Example: Student: ***“Left throttle identified”***

IP: ***“Left throttle verified”***

Student: ***“Left throttle idle, no change”***

Example: Student: ***“Left prop identified”***

IP: ***“Left prop verified”***

Student: ***“Left prop feather”***

“The Drill”

- Pitch for V_{YSE} (Blue line)
- Mixtures – FULL RICH
- Props – FULL FORWARD
- Throttles – FULL
- Flaps – UP (incrementally)
- Gear – UP
- Identify – *“Dead Foot, Dead Engine”*
- Verify – Retard throttle of inoperative (dead) engine to idle
- Feather – Feather prop of inoperative (dead) engine
- Zero-Sideslip – *“Raise the Dead, Split the Ball”*

ACS Tasks & Procedures

Normal Takeoff & Climb (Flaps 0°)

- Increase throttles to **2000 RPM**
- Check engine gauges
- Increase throttles to full power
- Rotate at V_R [**80 MPH** | **70 KIAS**]
- Climb at V_Y [**120 MPH** | **102 KIAS**]
- Establish two (2) positive rates of climb (Altimeter & VSI) then retract gear up
- At **1000'** AGL, reduce power to **24" MP** | **2400 RPM**
- Perform *Climb* checklist

Short Field Takeoff & Climb (Flaps 0°)

- Use all available runway and hold brakes though power up
- Increase throttles to **2000 RPM**
- Check engine gauges
- Increase throttles to full power and release brakes
- Rotate at V_R [**80 MPH** | **70 KIAS**]
- Climb at V_X [**107 MPH** | **89 KIAS**]
- Establish two (2) positive rates of climb (Altimeter & VSI) then retract gear up
- After clear of obstacle(s), accelerate to V_Y [**120 MPH** | **102 KIAS**]
- At **1000'** AGL, reduce power to **24" MP** | **2400 RPM**
- Perform *Climb* checklist

Steep Turns

- Perform clearing turns
- Pre-Maneuver Flow
 - **GUMPP** check (Gas **ON**, Undercarriage **UP**, Mixture **LEANED**, Props **2400 RPM**, Pumps **OFF**)
- Set power at **18-20" MP** and set heading bug on initial heading
- Maintain airspeed not exceeding **140 MPH | 120 KIAS**
- Roll into a coordinated 45° (50° commercial) bank turn
- Adjust power, pitch, and trim as necessary to maintain altitude and airspeed
- Maintain selected altitude +/- 100', selected airspeed +/- 10 KIAS, bank angle +/- 5°, and roll out on the entry heading +/- 10°
- Perform *Cruise* checklist

Slow Flight – configuration as specified by examiner

- Select entry altitude allowing task completion at no lower than 3,000' AGL
- Perform clearing turns
- Pre-Maneuver Flow
 - **GUMPP** check (Gas **ON**, Undercarriage **UP/DOWN**, Mixture **RICH**, Props **FULL**, Pumps **ON**)
- Reduce power to **12-15" MP** and smoothly extend flaps if needed to FULL below V_{FE}
- Configure as specified by examiner at the appropriate speeds
- Slow airspeed to stall warning horn, then increase airspeed by 5 MPH or 5 KIAS
- Adjust power as needed to maintain airspeed while maneuvering (approx. **20" MP**)
- Perform straight-and-level flight, climbs, turns, and descents as required
- Recover with max power and retract flaps to ½ if required
- Accelerate to V_X and retract landing gear if required
- Accelerate to V_Y and retract flaps to ¼, then UP if required
- Perform *Cruise* checklist

Power-Off Stalls – approach OR landing configuration

- Select entry altitude allowing task completion at no lower than 3,000' AGL
- Perform clearing turns
- Pre-Maneuver Flow
 - **GUMPP** check (Gas **ON**, Undercarriage **DOWN**, Mixture **RICH**, Props **FULL**, Pumps **ON**)
- Set power to **12-15" MP**, maintain altitude and slow to **100 MPH | 90 KIAS**
- Reduce power to **12" MP** and establish a stabilized descent
- Reduce power to **IDLE** while transitioning smoothly to flare pitch attitude after descent established
- Announce first indication of stall
- Recover at first indication or full stall as specified by the examiner
- Apply full power to recover
- Reduce AOA and level wings with minimum altitude loss
- Reduce flaps to ½ and establish climb at V_x
- Retract landing gear and accelerate to V_y
- Reduce flaps to 0°
- Perform *Cruise* checklist

Power-On Stalls – takeoff OR departure configuration

- Select entry altitude allowing task completion at no lower than 3,000" AGL
- Perform clearing turns
- Pre-Maneuver Flow
 - **GUMPP** check (Gas **ON**, Undercarriage **UP**, Mixture **RICH**, Props **FULL**, Pumps **ON**)
- Reduce power to **15" MP**, maintain altitude and slow to **80 MPH | 70 KIAS**
- Maintain selected heading or establish 15° bank turn, as specified by the examiner
- Transition smoothly to 15° pitch-up and increase power to **20" MP**.
- Announce first indication of stall
- Recover at first indication or full stall as specified by the examiner
- Apply full power to recover
- Reduce AOA and level wings with minimum altitude loss
- Establish climb at V_x
- Retract landing gear and accelerate to V_y
- Perform *Cruise* checklist

Accelerated Stalls

- Select entry altitude allowing task completion at no lower than 3,000" AGL
- Perform clearing turns
- Pre-Maneuver Flow
 - **GUMPP** check (Gas **ON**, Undercarriage **UP**, Mixture **RICH**, Props **FULL**, Pumps **ON**)
- Reduce power to **15" MP**, maintain altitude and slow to **120 MPH | 120 KIAS**
- Maintain selected altitude and establish a 45° bank turn
- Increase elevator back pressure smoothly and firmly to induce an impending stall
- Announce and recover at first indication of stall
- Apply full power to recover
- Reduce AOA and level wings with minimum altitude loss
- Establish climb at V_x
- Accelerate to V_y
- Perform *Cruise* checklist

V_{MC} Demo

- Select entry altitude allowing task completion at no lower than 3,000' AGL
- Perform clearing turns
- Pre-Maneuver Flow
 - **GUMPP** check (Gas **ON**, Undercarriage **UP**, Mixture **LEANED**, Props **2400**, Pumps **ON**)
- Reduce power to **15" MP** while maintaining heading and altitude and slowing to **120 MPH** or **120 KIAS**
- Simulate loss of critical engine by closing the **LEFT** throttle
 - **Perform "THE DRILL"** (up to, but not including "feather" – this maneuver is performed with windmilling prop)
- Upon reaching V_{YSE}, increase pitch attitude slowly, decreasing airspeed by 1 knot/second until full rudder is required to maintain directional control
- Recover at first sign of:
 - Loss of directional control, OR
 - First indication of stall (stall horn/buffet/decay of control effectiveness) ***MOST LIKELY***
- Recover by reducing power on right engine while simultaneously decreasing AOA to regain directional control
- Increase power slowly on right engine and pitch for V_{SSE} or V_{YSE} as specified by the examiner
- Recover by bringing throttles slowly together
- Perform *Cruise* checklist

Drag Demonstration

- Select entry altitude allowing task completion at no lower than 3,000' AGL
- Perform clearing turns
- Pre-Maneuver Flow
 - **GUMPP** check (Gas **ON**, Undercarriage **UP**, Mixture **LEANED**, Props **2400**, Pumps **ON**)
- Simulate loss of engine by closing one of the throttles
 - **Perform "THE DRILL"** up to but not including "Feather" – *Note VSI*
 - **Finish "THE DRILL"** – Simulated feather is **11-12" MP** on the "inoperative" engine – *Note VSI*
- Enter zero-sideslip by banking up to 5° into right engine and splitting the ball – *Note VSI*
- Lower landing gear – *Note VSI*
 - Raise landing gear
- Increase flaps incrementally to full – *Note VSI*
- Lower landing gear – *Note VSI (This is the drag worst case scenario)*
- Reverse the procedure by raising the flaps, then gear, then final flaps until clean
- Bring throttles slowly together
- Perform *Cruise* checklist

Normal Approach & Landing

- Complete *Descent* checklist before entering TPA or for straight-in
- Slow to **120 MPH | 120 KIAS** prior to entering traffic pattern (**18" MP | 2400 RPM**)
- Enter traffic pattern at published TPA
- Abeam landing point on the downwind leg: **15" MP**, extend flaps $\frac{1}{4}$, and lower landing gear
- **1st GUMPP** check (**Gas ON, Undercarriage DOWN, Mixture FULL, Props 2400 RPM, Pumps ON**)
- While or after turning base leg, extend flaps to $\frac{1}{2}$
- Base leg at **110 MPH | 110 KIAS**
- **2nd GUMPP** check (**Gas ON, Undercarriage DOWN, Mixture FULL, Props 2400 RPM, Pumps ON**)
- While or after turning to final, extend flaps to FULL and maintain
- **FINAL GUMPP** check (**Gas ON, Undercarriage DOWN, Mixture FULL, Props FULL, Pumps ON**)
- Roll out on final at approximately **100 MPH | 100 KIAS**
- Barrier speed should be approximately **85-90 MPH/KIAS (E-Model or non-STOL F-Model) | 75-80 KIAS (STOL F-Model)**
- Slowly retard throttles and touchdown at approximately **80 MPH | 70 KIAS**
- Trim sufficiently and maintain back pressure to prevent nose wheel damage

Short Field Approach & Landing

- Complete *Descent* checklist before entering TPA or for straight-in
- Slow to **120 MPH | 120 KIAS** prior to entering traffic pattern (**18" MP | 2400 RPM**)
- Enter traffic pattern at published TPA
- Abeam landing point on the downwind leg: **15" MP**, extend flaps $\frac{1}{4}$, and lower landing gear
- **1st GUMPP** check (**Gas ON, Undercarriage DOWN, Mixture FULL, Props 2400 RPM, Pumps ON**)
- While or after turning base leg, extend flaps to $\frac{1}{2}$
- Base leg at **110 MPH | 110 KIAS**
- **2nd GUMPP** check (**Gas ON, Undercarriage DOWN, Mixture FULL, Props 2400 RPM, Pumps ON**)
- While or after turning to final, extend flaps to FULL and maintain
- **FINAL GUMPP** check (**Gas ON, Undercarriage DOWN, Mixture FULL, Props FULL, Pumps ON**)
- Roll out on final at approximately **100 MPH | 100 KIAS**
- Barrier speed should be approximately **85-90 MPH/KIAS (E-Model or non-STOL F-Model) | 75-80 KIAS (STOL F-Model)**
- Slowly retard throttles and touchdown at approximately **80 MPH | 70 KIAS**
- Trim sufficiently and maintain back pressure to prevent nose wheel damage

Touch and Go Operations (No less than 5,000' runway)

- After touchdown: IP will guard the gear handle
- Student must identify the flap handle before flap retraction
- Once flaps are up and verified and trim is set for takeoff, smoothly increase throttles to full power.
- Rotate at V_R [**80 MPH | 70 KIAS**]
- Climb at V_Y [**120 MPH | 102 KIAS**]
- Establish two (2) positive rates of climb (Altimeter & VSI) then retract gear up
- At **1000' AGL**, reduce power to **24" MP | 2400 RPM**
- Perform *Climb* checklist
- For pattern operations:
 - See Normal Approach & Landings or Short Field Approach & Landings

Go-Around/Balked Landing

- When decision made to abort landing and “go-around”, increase throttles to full power
- Retract flaps to ½ and pitch as necessary to accelerate to V_X
- Establish V_X climb with two positive rates then retract landing gear
- Accelerate to V_Y then retract remaining flaps incrementally
- At TPA, reduce power to **24” MP | 2400 RPM**
- If departing the pattern (enroute, instrument navigation, vectors, etc.)
 - Climb at **120 MPH | 102 KIAS** after 1000’ AGL
- If remaining in the pattern
 - Maintain **120 MPH | 120 KIAS (18” MP | 2400 RPM)** at TPA

Instrument Approach Procedures (Precision & Non-Precision)

- Complete *Approach* briefing
 - Set up radios and flight instruments
- Complete *Descent* checklist
- Establish 120 MPH | 120 KIAS until GS intercept (precision) or FAF inbound (non-precision)
 - **18”-20” MP | 2400 RPM**
- At GS intercept (precision) or FAF (inbound), reduce to **18”-20” MP**
 - **Flaps ¼, Gear Down, Start Down, Start Time, Call Tower**
 - If OEI, wait to extend flaps and landing gear until a minimum of V_{YSE} can be maintained
- **GUMPP** check (**Gas ON, Undercarriage DOWN, Mixture FULL, Props FULL, Pumps ON**)
- Continue landing with flaps as required or execute missed approach as specified by the examiner

Emergency Procedures

Takeoff Emergencies

Engine Failure Before V_R

- Abort takeoff – close throttles
- Maintain directional control
- Brake as required to stop on runway
 - If insufficient runway remains, mixture IDLE CUTOFF
 - Fuel selectors, magnetos, and Master OFF
 - Avoid obstacles and stop straight ahead

Engine Failure After V_R (Gear Down) – *usable runway distance available*

- Abort takeoff – close throttles
- Maintain directional control
- Pitch down to increase airspeed
- Land straight ahead
- Brake as required to stop on runway
 - If insufficient runway remains, mixture IDLE CUTOFF
 - Fuel selectors, magnetos, and Master OFF
 - Avoid obstacles and stop straight ahead
 - Secure cabin if insufficient runway remains

Engine Failure After V_R (Gear Up) – *out of usable runway distance*

- **Execute “The Drill”**
- Check VSI
 - Positive rate of climb – continue climb and circle to land
 - Negative rate of climb – land straight ahead while avoiding obstacles
- Declare the emergency

In Flight Emergencies

In-Flight Engine Failure (Below 3000' AGL)

- **Execute “The Drill”**
- Perform *Engine Shutdown/Secure (Air)* Checklist
- Declare the emergency

In-Flight Engine Failure (Above 3000' AGL)

- **Execute “THE DRILL”** up to but not including “Feather”
 - Perform *Troubleshoot* Checklist
 - If engine restarts, land as soon as practicable
 - If engine fails to restart, continue emergency
 - **Finish “THE DRILL”**
- Perform *Engine Shutdown/Secure (Air)* Checklist
- Declare the emergency

Emergency Descent

- Close throttles
- Propellers full forward
- Mixtures adjust
- Cowl flaps as required
- Extend gear once below V_{LE} and maintain **140 MPH | 120 KIAS**
- Use 30°-45° of bank to maintain positive load factors during descent
- Notify ATC as appropriate

Unintentional Spin Recovery –SPINS ARE NOT AUTHORIZED IN THE AZTEC

- **Power** (throttle) to idle
- **Ailerons** neutral
- **Rudder** applied in opposite direction of spin (“*step on the high wing*”)
- **Elevator** (back pressure) reduced, then full forward to break stall
- Neutralize rudder when rotation stops
- Apply smooth back pressure to recover from dive

Recovery from Unusual Attitudes

- **Nose High Climbing** (*Airspeed Decreasing*)
 - Increase power
 - Pitch level
 - Bank 0°
 - Perform *Cruise* checklist
- **Nose Low Descending** (*Airspeed Increasing*)
 - Reduce power
 - Bank 0°
 - Pitch level
 - Perform *Cruise* checklist

